

Estimates of the reproduction ratio from epidemic surveillance may be biased in spatially structured populations

About the bias from non-spatially explicit $R(t)$ estimation. Does not derive NGM etc, only one type of reproduction number. Sets up our problem nicely as justifies why bother. Convergence to equilibrium similar

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Accurate estimates of the reproduction ratio are crucial for projecting the evolution of an infectious disease epidemic and for guiding the public health response. Here we prove that estimates of the reproduction ratio based on inference from surveillance data can be inaccurate if the population comprises spatially distinct communities, as the space–mobility interplay may hide the true evolution of the epidemic from surveillance data. Consequently, surveillance may underestimate the reproduction ratio over long periods, even mistaking growing epidemics as subsiding. To address this, we use the spectral properties of the matrix describing the spatial epidemic spread to reweight surveillance data. We propose a correction that removes the bias across all epidemic phases. We validate this correction against simulated epidemics and use COVID-19 as a case study. However, our results apply to any epidemic in which mobility is a driver of circulation. Our findings may help improve epidemic monitoring and surveillance and inform strategies for public health responses.

The reproduction ratio R is arguably the indicator most used for monitoring the trend in the evolution of an infectious disease epidemic. R is the average number of secondary infections that each case generates. When it is larger than one, the epidemic wave is growing. When it is lower than one, it is subsiding^{1,2}. The reproduction ratio also measures the effectiveness of public health interventions, whose overarching goal is to bring an unconstrained epidemic ($R > 1$) below the epidemic threshold of $R = 1$. Accurately estimating the reproduction ratio is, thus, necessary for ascertaining the current epidemic evolution, for predicting short-term trends, for scenario analyses and for planning public health action^{3–7}. The standard way to measure R is to infer it from data from epidemiological surveillance^{8–12}. These data may be timelines of detected cases or their proxies, like hospitalizations or deaths, and this approach applies to diseases spanning radically different epidemiologies, transmission routes and burdens, like influenza^{13,14}, measles¹⁵, COVID-19 (ref. 16), Ebola¹⁷, cholera¹⁸, dengue¹⁹ and malaria²⁰. The resulting surveillance-based estimates of R are routinely used to design interventions²¹.

Notwithstanding this usefulness, we argue in this study that surveillance data may lead to biased estimates of the reproduction ratio in spatially structured populations in which geographically distinct communities (for example, cities) are connected through human mobility. Heterogeneities in the contact network shape the way epidemics spread²². Here, we will show that those arising from the complex interplay between spatial transmissibility patterns and the mixing network driven by human mobility hide the true dynamic structure of the epidemic process from population-level surveillance data. This mirrors the nature of most mathematical models of epidemic spread. These integrate space and spatial data at high resolution^{22–27}, but they find it harder to do the reverse, which is to extract high-resolution information from limited and coarse-grained surveillance data in the absence of any knowledge of the underlying spatial dynamics^{28–30}. Crucially, this means that inference on surveillance data may either overestimate or underestimate R over long periods. This is of great public health relevance, as measuring, for instance, a reproduction

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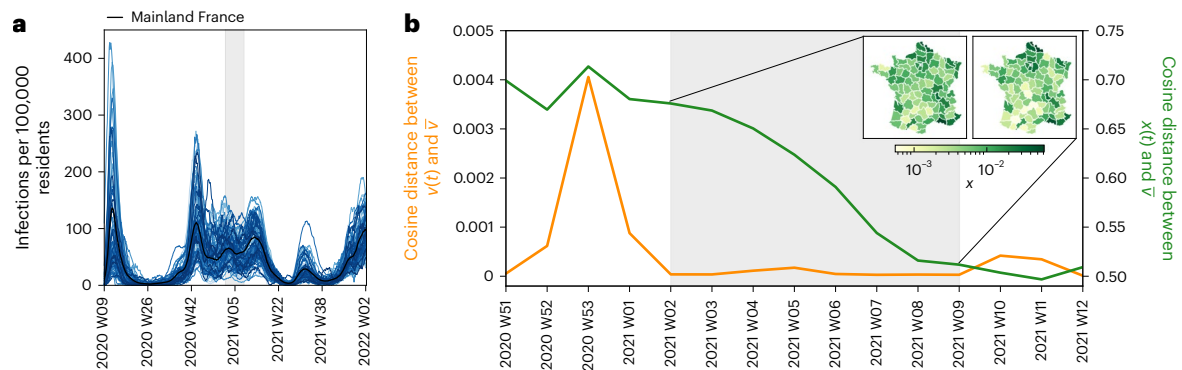


Fig. 1 | Convergence to the equilibrium spatial distribution of infections.

a, Estimate of the daily number of new COVID-19-related infections per 100,000 residents from week 9 of 2020 to week 3 of 2022 (23 February 2020 to 23 January 2022). See Supplementary Section 1.3 for how we estimated infections from hospitalizations and for a comparison with serological surveys. Each curve is for one of the 94 departments (administrative level 2) of mainland France, excluding the region of Corsica, with a 14 d window rolling average. The period from week 2 to week 9 of 2021 (11 January to 7 March), selected as a period in which $\mathbf{v}$ was fairly constant, is highlighted in grey in both **a** and **b**. The small department Territoire de Belfort (0.2% of the French population) is excluded due to a miscounting of

hospitalizations from neighbouring departments by the health authorities (Supplementary Figs. 3 and 4). **b**, The orange curve (left y axis) reports the weekly cosine distance between $\mathbf{v}$ (computed from mobility data from that week; see section ‘Reconstruction of the reproduction operator from data’) and its average ($\bar{\mathbf{v}}$) from week 51 of 2020 to week 12 of 2021 (Supplementary Fig. 13). The cosine distance between $\mathbf{v}$ and $\bar{\mathbf{v}}$ is defined as $1 - \mathbf{v} \cdot \bar{\mathbf{v}} / (\|\mathbf{v}\| \|\bar{\mathbf{v}}\|)$. The green curve (right y axis) reports the cosine distance between $\mathbf{v}$ and the estimated weekly spatial distribution of COVID-19 infections $\mathbf{x}$ in French departments. The two inset maps show the spatial distribution $\mathbf{x}$ in week 2 and week 9 of 2021, respectively (edges of the time interval in grey).

ratio below one when the true value is above would falsely signal that the epidemic is under control.

Here, we study this bias, identify its origin and compute its magnitude. Then, we propose a correction to case incidence data that removes this bias and ensures that surveillance-based estimates of the reproduction ratio consistently give the true value for the epidemic. Our theoretical findings apply to any epidemic with a relatively short generation time and for which mobility is a contributing factor in shaping its circulation within and across communities. This covers some of the global health threats that are worst affected by climate change and demographic trends: viruses responsible for respiratory infections (including SARS-CoV-2 and influenza)³¹, vector-borne pathogens (including the arboviruses dengue, chikungunya and Zika)^{32,33} and the emergence of new viruses or new viral strains³⁴. To test and illustrate our findings, we use the French COVID-19 epidemic (Fig. 1) before the advent of vaccination as a case study.

Theoretical formalism

The Galton–Watson branching process is a customary framework for modelling epidemic spread^{35–37}. Let $I(0)$ be the initial number of infections, $I(1)$ the expected number of infections that the initial cases generate and, generally, let $I(t)$ be the expected number of infections in the t th generation. By definition of the reproduction ratio, we have that $I(t) = R I(t-1)$, which implies that $I(t) = R^t I(0)$. This equation means that the number of infections grows exponentially if $R > 1$. In any real outbreak, other factors, like acquired immunity, seasonal effects or public health interventions, will at some point curb this exponential growth by changing the value of R . Notwithstanding, we may assume R to be fairly constant, either in the early phase of an outbreak when those effects have not yet kicked in or when the timescale at which immunity and mixing change is much longer than that of epidemic evolution^{38,39}.

For a population composed of N spatial communities, we may define the vector $\mathbf{I}(t) \in \mathbb{R}^N$, whose component $I(t)_i$ is the number of infections in generation t and community i . Likewise, the reproduction operator $R \in \mathbb{R}^{N,N}$ encodes in component R_{ij} the average number of infections generated among the residents of community i by a case belonging to community j (ref. 40). Its definition, thus, matches that of a spatially structured next-generation matrix^{41,42}. However, unlike

the standard next-generation matrix, it can be built at any phase of the epidemic and not only close to the disease-free state, and there is no need to assume a specific transmission model. The definition of R and the results that we are going to derive from it, thus, apply to any epidemic and disease. The specific parametrization of R will instead depend on the specific transmission dynamics and natural history of the disease. For directly transmitted diseases, R typically depends on mixing patterns among communities⁴³. For vector-borne diseases, the local abundance of the host vectors, which modulate the effective transmissibility, needs to be factored in, too^{32,44}. The expected epidemic evolution then follows the equation

$$\mathbf{I}(t) = R^t \mathbf{I}(0). \quad (1)$$

$\mathbf{I}(t)$ encodes both the total number of infections in the population in generation t and its spatial distribution. We define the former as the number $I_{\text{tot}}(t) = \sum_i I(t)_i$ and the latter as the vector $\mathbf{x}(t) \in \mathbb{R}^N$ whose components are $x(t)_i = I(t)_i / I_{\text{tot}}(t)$.

The reproduction ratio of this process is the eigenvalue of R with the largest absolute value (spectral radius)^{41,42}, which is real, positive and nondegenerate (Supplementary Section 1.1). Its corresponding eigenvector $\mathbf{v}$ is also strictly positive ($v_i > 0$), and we normalize it so that $\sum_i v_i = 1$.

Measuring the reproduction ratio of the system thus requires knowledge of the spectral structure of R , that is, of the spatial structure of the epidemic. We call this the reference reproduction ratio R^{ref} as opposed to what surveillance instead estimates from the evolution of the incidence of infections or their proxies. This may happen globally, at the level of the entire population or locally in each community. In our framework, the population-level estimated reproduction ratio is $R^{\text{estim}}(t) = I_{\text{tot}}(t+1)/I_{\text{tot}}(t)$, that is, the generational growth rate. The local community-level estimated reproduction ratio is instead $r_i^{\text{estim}}(t) = I(t+1)_i / I(t)_i$.

A simple observation then underpins our study. In general $R^{\text{estim}}(t)$ and $r_i^{\text{estim}}(t)$ may be different from R^{ref} , the spectral radius of R , and, if that is the case, surveillance will not estimate the reference reproduction ratio.

To explore this, we will first determine the conditions leading to an unbiased measure of the reproduction ratio: $R^{\text{estim}}(t) = R^{\text{ref}}$.

When reference and estimated reproduction ratios match

By virtue of the Perron–Frobenius theorem, $R^t \rightarrow R^{\text{ref}} \mathbf{v} \mathbf{v}^*$ asymptotically at large t , where $\mathbf{v}^*$ is the dual of $\mathbf{v}$ (and left Perron eigenvector of R) and normalized so that $\mathbf{v}^* \mathbf{v} = 1$. Equation (1) becomes $\mathbf{I}(t) \rightarrow [\mathbf{v}^* \mathbf{I}(0)] R^{\text{ref}} \mathbf{v}$, which implies that $\mathbf{x}(t) \rightarrow \mathbf{v}$, $R^{\text{estim}}(t) \rightarrow R^{\text{ref}}$ and $r_i^{\text{estim}}(t) \rightarrow R^{\text{ref}}$. The epidemic dynamics, thus, brings the spatial distribution of infections towards $\mathbf{v}$, which we will refer to as the equilibrium spatial distribution of infections. Thus, for any epidemic dynamics, if infections are spatially distributed as the equilibrium distribution ($\mathbf{x} = \mathbf{v}$), then the error is zero and the reference reproduction ratio is measured both globally ($R^{\text{estim}} = R^{\text{ref}}$) and locally ($r_i^{\text{estim}} = R^{\text{ref}}$). The last equality implies the existence of a detailed balance between locally generated, imported and exported infections in every community (Supplementary Section 1.2). Figure 1 shows evidence of the convergence to $\mathbf{v}$ during the COVID-19 epidemic in France in late 2020 and early 2021. We reconstructed R from mobility data from Meta⁴⁵, a multinational technology company, and $\mathbf{x}$ from hospitalization data (Fig. 1, section ‘Reconstruction of the reproduction operator from data’ and Supplementary Section 1.3). In a period when R was fairly constant, the angle between $\mathbf{x}$ and $\mathbf{v}$ consistently decreased, a sign that $\mathbf{x}$ was evolving towards the equilibrium distribution, as the theory predicts.

Finally, there is a class of operators R for which the reference and population-level estimated reproduction ratios match at all times, even out of equilibrium. Indeed, if the expected total number of secondary infections $\rho_i \equiv \sum_j R_{ji}$ is the same across communities, then $R^{\text{estim}} = \rho_i = R^{\text{ref}}$, $\forall i$ identically (section ‘Homogeneous transmission potential’). This implies that only the combination of spatial epidemic coupling and spatial heterogeneity in the transmission potential ρ_i may cause R^{estim} to be different from R^{ref} . Notably, this is true only globally and r_i^{estim} may be different from R^{ref} even if $\rho_i = R^{\text{ref}}$.

When reference and estimated reproduction ratios do not match

We now focus on the out-of-equilibrium dynamics ($\mathbf{x}(t) \neq \mathbf{v}$) and measure the bias on R^{estim} as the relative difference between the estimated and the reference reproduction ratios:

$$\Delta(t) = \frac{R^{\text{estim}}(t) - R^{\text{ref}}}{R^{\text{ref}}}. \quad (2)$$

We call Λ_α ($\alpha = 1, \dots, N-1$) the (possibly degenerate) eigenvalues of R other than R^{ref} and, by the Perron–Frobenius theorem, $|\Lambda_\alpha| < R^{\text{ref}}$. With the calculations reported in the section ‘Proof of equation (3)’, we find that

$$\Delta(t) = C(t) \sum_\alpha z_\alpha \left(1 - \frac{\Lambda_\alpha}{R^{\text{ref}}} \right) \left(\frac{\Lambda_\alpha}{R^{\text{ref}}} \right)^t, \quad (3)$$

where $C(t)$ is positive and asymptotically constant, and z_α is a (possibly complex) number proportional to the projection of the initial condition $\mathbf{x}(0)$ on the α th mode. The modes in equation (3) for which $\Lambda_\alpha \approx R^{\text{ref}}$, or that are almost orthogonal to the initial configuration $\mathbf{x}(0)$, are suppressed from the start and do not bias the estimate of the reproduction ratio. The other modes, instead, possibly do, with an effect that becomes smaller with a characteristic decay time $\tau_\alpha = 1 / \log(R^{\text{ref}}/|\Lambda_\alpha|)$. In addition, those modes for which Λ_α is not real and positive oscillate with period $T_\alpha = 2\pi/|\theta_\alpha|$, where $\theta_\alpha = \arg \Lambda_\alpha$ (with $\theta_\alpha \in (-\pi, \pi]$); see section ‘Calculation of τ_α and T_α ’. These oscillations will be visible if they are faster than their characteristic decay time, $T_\alpha \leq \tau_\alpha$. This gives the inequality:

$$\frac{|\Lambda_\alpha|}{R^{\text{ref}}} \geq e^{-\frac{|\theta_\alpha|}{2\pi}} \geq e^{-\frac{1}{2}} \approx 0.61. \quad (4)$$

The strict equality occurs when Λ_α is real and negative ($\theta_\alpha = \pi$). Furthermore, using the same approach, we prove in Supplementary Section

1.2 that oscillations may be visible in the local estimated reproduction ratios even when the global $\Delta(t)$ never changes sign (Fig. 2e).

To test the predictions of our theory in a realistic scenario, we considered again the COVID-19 epidemic in France and built a stochastic metapopulation model using the same data as in Fig. 1b. The model is described in detail in section ‘Epidemic simulations’ and Supplementary Section 1.4. We measured the reference and the estimated reproduction ratios, as reported in Fig. 2, which shows that surveillance-based estimates may remain consistently biased for a long period and, depending on where the epidemic wave started, they may either overestimate or underestimate the reference reproduction ratio (see Supplementary Fig. 9 for more initial conditions tested). The case depicted in Fig. 2d is particularly concerning. During the first month of the simulated epidemic, surveillance records a lower-than-one reproduction ratio, which indicates decreasing incidence (visible in Fig. 2b), mistakenly pointing to a subsiding outbreak. In reality, the reproduction ratio is well above one, and only after two months of the simulated epidemic does the surveillance-based estimate reach this value. Alongside the estimate of R^{estim} given within the framework of the Galton–Watson process, Fig. 2c,d also provides estimates of the reproduction ratio by feeding incident cases to the library EpiEstim¹¹, a popular tool for computing the reproduction ratio from surveillance data. That the two measures overlap confirms that the Galton–Watson process correctly reproduces the phenomenology under study, even in realistic scenarios. Notwithstanding, more detailed frameworks^{41,46,47} could be used to study the impact of heterogeneous generation intervals.

Finally, Fig. 2e,f shows that locally measured reproduction ratios converge to the reference value at different times and with different speeds, and that, at the same moment in time, some communities overestimate R^{ref} and some underestimate it (see the analytical proof in section ‘Local over- and underestimation of R^{ref} ’).

Figure 2 shows no oscillations in the sign of $\Delta(t)$, compatible with the reconstructed operator R having only real and positive eigenvalues. We extended our analysis to 32 European countries and none showed oscillations in $\Delta(t)$ at the considered spatial resolution (admin-2). We discuss the details of this analysis and the possible realistic conditions leading to oscillations in Supplementary Section 1.5 and Supplementary Fig. 1.

In the absence of oscillations, the estimated reproduction ratio consistently either overestimates or underestimates the reference reproduction ratio, as $\Delta(t)$ decays to zero without ever changing sign. In this case, we can determine the sign of the bias from the initial condition: $\Delta(0) = \sum_j \rho_j x(0)_j - R^{\text{ref}}$. By the Perron–Frobenius theorem, $j_{\min}$ and $j_{\max}$ exist, so that $\rho_{j_{\min}} \leq R^{\text{ref}}$ and $\rho_{j_{\max}} \geq R^{\text{ref}}$. Thus, the initial location of infections will completely determine the sign of the error that surveillance will make. If the epidemic starts at $j_{\max}$ (or, in general, in communities with a high transmission potential), surveillance will consistently overestimate the reference reproduction ratio until the bias decays to zero. Conversely, if it starts at $j_{\min}$ (or in communities with low transmission potential), surveillance will underestimate R^{ref} . The spatial structure of the epidemic, thus, generates something like a core-group effect, an epidemiological concept whereby the disease is initially confined to specific groups for which strong clustering, high susceptibility or specific behavioural traits result in transmission rates higher than in the general population (like, for instance, schoolchildren and influenza⁴⁸) and requires specific corrections to epidemic models^{42,49–51}. In our case, areas with higher-than-average ρ_i play the role of core groups, skewing the estimated reproduction ratio to values higher than R^{ref} . We restate, however, that if the epidemic is, instead, localized in areas with lower-than-average ρ_i , the effect is the opposite and leads to an underestimate of R^{ref} , something that is not typically considered in the standard core-group theory. Moreover, just as core-group effects may make it harder to assess and predict the population-level impact of an epidemic, the bias that the spatial structure induces on the

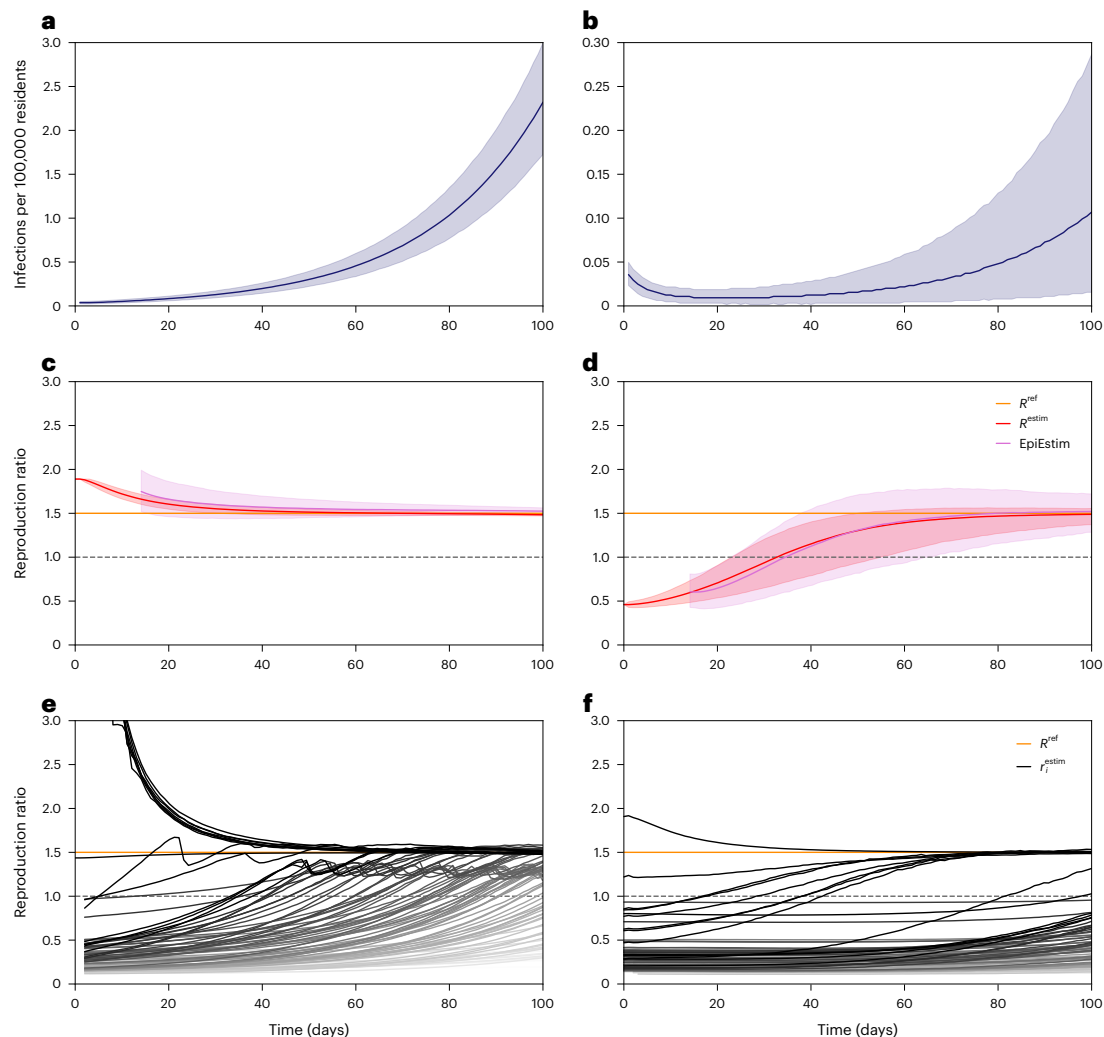


Fig. 2 | Comparison of the reference and measured reproduction ratios. a–f, We generated stochastic simulations of a COVID-19-like epidemic in France using a metapopulation model, whereby we divided the French population into 94 spatial communities corresponding to the departments of mainland France minus Corsica, and modelled within- and between-community mixing using mobility data from Meta. See section ‘Epidemic simulations’ and Supplementary Section 1.4 for details on how we built R from mobility data and on the epidemic spread model. **a,b,** Daily number of incident infections per 100,000 individuals for 100 initial infections seeded in the department of Paris (**a**) or among departments in proportion to their population (**b**). **c,d,** R^{ref} (yellow) is the

reference reproduction ratio, R^{estim} (red) is the estimated reproduction ratio as described in this paper and EpiEstim (purple) is the reproduction ratio computed from incidence data using the EpiEstim package¹¹ for 100 initial infections seeded in the department of Paris (**c**) or among departments in proportion to their population (**d**). **e,f,** r_i^{estim} is the locally estimated reproduction ratio for 100 initial infections seeded in the department of Paris (**e**) or among departments in proportion to their population (**f**). See Supplementary Section 1.4 for details. We used 10,000 runs of the stochastic model. The medians are plotted as solid lines and the 95% confidence intervals as shaded areas. For r_i^{estim} , confidence intervals are omitted for readability.

reproduction ratio prevents an accurate and timely estimate of the epidemic trajectory, possibly misinforming the public health response. The next section will present a method to overcome this.

Note, however, that, at present, our formalism covers only core-group effects related to the spatial structure. Other drivers, like the presence of clusters in the contact network, require additional complexity, such as featuring an initial subexponential growth of infections⁵⁰ to including the explicit contact network⁵¹ inside each spatial community. Both can be future extensions of the model presented here, but they require changing the functional form of equation (1) in the former case or forgoing the analytical approach altogether in the latter case. Another known source of bias on the reproduction ratio is the importation of infections from outside the system, for example, from other countries⁵². Corrections exist for an initial overestimation due to miscounting imported infections as infections generated within the systems, and these could be used in combination with our correction⁵³. Our correction, however, also addresses the impact that

imported infections may have in changing the spatial distribution of the epidemic. Indeed if importations are distributed differently from the equilibrium distribution $\mathbf{v}$, and there is no reason why they should, they will enhance the bias on the reproduction ratio arising from $\mathbf{x}$ being different from $\mathbf{v}$, which is the object of our study.

Correction to surveillance data

So far we have proven that surveillance-based estimates of the reproduction ratio may be biased. We will now propose a way to correct for this bias. The global incidence of infections that is used to estimate the reproduction ratio is trivially proportional to the unweighted average of the incidence across communities: $I_{\text{tot}}(t) = N(\sum_i I_i(t)/N)$. We now define a modified incidence that weights local infections with the Perron dual vector:

$$I_{\text{tot}}^{(v)}(t) = N \left(\frac{\sum_i v_i^* I_i(t)}{\sum_i v_i^*} \right) = N \sum_i v_i^* I_i(t) = N \mathbf{v}^* \mathbf{I}(t). \quad (5)$$

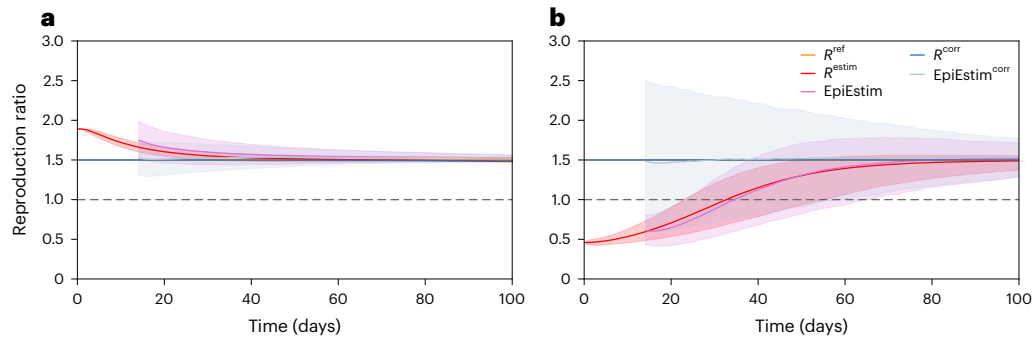


Fig. 3 | Corrected reproduction ratio. a, b. This figure uses the same stochastic epidemic model as in Fig. 2 and the same initial conditions, that is, **a** and **b** correspond to **c** and **d** in Fig. 2 respectively. Medians and 95% confidence intervals are computed and shown over 10,000 runs. We compare the standard

reproduction ratio measured from surveillance data, for both the Galton–Watson formalism (red) and EpiEstim (purple) on incidence data, and the corresponding $\mathbf{v}^*$ -corrected estimates: Galton–Watson formalism (blue) and with EpiEstim (light blue) on modified incidence data $I_{\text{tot}}^{(v)}$.

Then, we use it to get a new estimate of the reproduction ratio:

$$R_{\text{corr}}(t) = \frac{I_{\text{tot}}^{(v)}(t+1)}{I_{\text{tot}}^{(v)}(t)} = \frac{\mathbf{v}^* \mathbf{I}(t+1)}{\mathbf{v}^* \mathbf{I}(t)} = \frac{\mathbf{v}^* R \mathbf{I}(t)}{\mathbf{v}^* \mathbf{I}(t)} = R_{\text{ref}} \frac{\mathbf{v}^* \mathbf{I}(t)}{\mathbf{v}^* \mathbf{I}(t)} = R_{\text{ref}}, \quad (6)$$

and find it is identically equal to the reference value. The practical advantage for epidemic monitoring is clear: our correction gives an unbiased estimate of the reproduction ratio from surveillance data all along the epidemic wave, unlike traditional measures. It has, however, two potential drawbacks. The former is that if the outbreak starts in communities where I_i^* is small, then $\mathbf{v}^* \mathbf{I}(t)$ will be very small. Stochastic fluctuations would then cause large changes in R_{corr} . In that case then, R_{corr} may well be accurate but not precise. However, no initial condition can be orthogonal to $\mathbf{v}^*$ whose entries are strictly positive, so even if $\mathbf{v}^* \mathbf{x}$ is initially small, it would probably increase quickly and with it the precision of the measurement. Figure 3 shows that R_{corr} accurately measures the reference reproduction ratio from the beginning of the epidemic wave for the simulated epidemics in Fig. 2. Notably, Fig. 3 also shows that if you feed $I_{\text{tot}}^{(v)}(t)$ into EpiEstim instead of $I_{\text{tot}}(t)$, you will also completely remove the bias on the estimate of the reproduction ratio. Our proposed modified incidence can then be readily incorporated into standard tools for public health surveillance to improve their accuracy.

The latter potential drawback is that our correction requires knowing $\mathbf{v}^*$. We argue, however, that this does not require knowing or measuring R in real time and that a good estimate of $\mathbf{v}^*$ for epidemic monitoring can be computed during peacetime from past population and mobility data (pre-epidemic or from data collected during earlier epidemic phases). Indeed, $\mathbf{v}^*$ is more stable than R because any change occurring homogeneously across communities (for example, changes in the rate of immunity or public health interventions) would change the latter, not the former. Figure 4 compares the standard estimated reproduction ratio of COVID-19 in France between late 2020 and March 2021 to our correction. The former is computed with EpiEstim on inferred case incidence; the latter is computed with EpiEstim on the corrected incidence $I_{\text{tot}}^{(v)}$, with $\mathbf{v}^*$ computed from various samples of past mobility data, all yielding stable estimates. Our correction suggests that traditional surveillance underestimated the reproduction ratio of COVID-19 in France during January and February 2021. Notably, surveillance recorded a lower-than-one reproduction ratio for more than 2 weeks (see also the official reports from that time⁵⁴), indicating a subsiding epidemic wave. This is at odds with what we know happened: a growing epidemic wave, namely the third French wave. This led to the national lockdown enforced on 3 April 2021, which was immediately after the time window depicted in Fig. 4. By correctly weighting the infections in departments with low transmission potential, our

corrected reproduction ratio would have, instead, consistently signalled a growing epidemic wave throughout the first three months of 2021 (Supplementary Figs. 12 and 13). This discrepancy is notable in the context of the debate over the public health response at that time. In early 2021, a national curfew was in effect but cases were rising due to the introduction and gradual takeover of the Alpha variant of SARS-CoV-2. The authorities were wary of additional restrictions and were relying on mass vaccination despite models suggesting that that might not be enough⁵⁵, as only 3% of the population had received at least one dose by mid-February⁵⁴ (week 6 of 2021 in Fig. 4). It is conceivable, albeit circumstantial, that the underestimation of the severity of the wave by surveillance could have contributed to delaying the enforcement of stricter movement restrictions, which anyway became inevitable later in April.

Note that our method removes a source of bias that it is inherent to the spatial dynamics of the system and may be present even when surveillance data are accurate, representative and unbiased. In this sense, it should integrate methodologies that address other sources of bias such as uneven susceptibility and case detection rate, reporting delays, inaccurate reconstruction of infections from proxy timeseries and inaccurate estimates of the generation time^{51,56–59}. We prove the computational feasibility of this integration in our library⁶⁰. It is also important to stress, however, that our correction is as good as the data used to calculate it, and specifically the data used to reconstruct R . The different global health threats we identified in the introduction entail different minimum requirements on the data. COVID-19 and influenza need data on local and between-community mixing, whereas arboviral diseases (dengue, chikungunya and Zika) need data on where residents of each community spend their time and the relative spatial distribution of the vector population (*Aedes* mosquitoes). Finally, spreading routes and disease natural history will inform the specific data needs in the case of newly emerging pathogens or strains. Overall, however, large-scale estimates of human mobility and mixing will be necessary, which increasingly come from mobile phone data. The impact of their own limitations^{61–65} will require a case-by-case assessment when these data are used to compute our correction. In our case study, Fig. 4 shows that the correction is robust to choosing human mixing data from different time periods. Also, both the data provider and we implemented adjustments to improve the accuracy and representativeness of the colocation data⁴⁵ (section ‘Reconstruction of the reproduction operator from data’). Finally, our correction is currently static and works over periods of fairly constant R . Future studies should address how $\mathbf{v}^*$ should change in time to consistently correct for the bias on the reproduction ratio by explicitly coupling population mixing and disease natural history and handling exogenous shifts in R due to seasonal trends, behavioural changes and interventions.

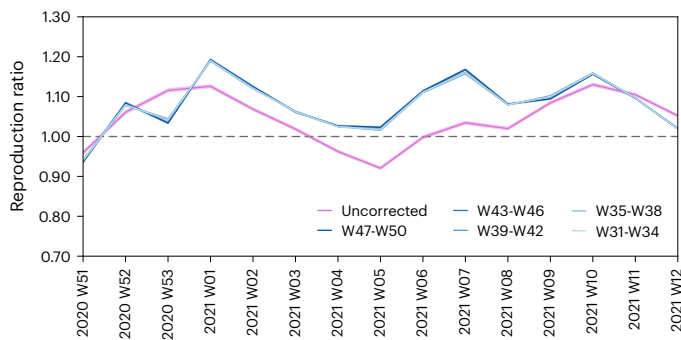


Fig. 4 | Application of the proposed correction to COVID-19 data for France.

The pink curve is the reproduction ratio of COVID-19 in France obtained from country-level incidence, in the same time interval as in Fig. 1b, computed using the package EpiEstim. The blue curves are the corrected reproduction ratio obtained by feeding EpiEstim the corrected incidence, as in equation (5). For this, the Perron dual vector $\mathbf{v}^*$ is reconstructed from mobility data at different times before the observation period, as specified in the legend as week ranges of 2020. The shaded areas are the 95% credibility intervals computed by EpiEstim, which are shown for readability only for the pink curve and the darkest blue curve. The different blue curves are hardly distinguishable because they overlap. See section ‘Reconstruction of the reproduction operator from data’ for a detailed explanation of how to compute $\mathbf{v}^*$ from data.

Our study describes a practicable way to improve the accuracy of the information that flows from epidemiological surveillance to public health policymakers. Better information may lead to more effective policies for preventing and controlling epidemic threats.

Online content

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Methods

Homogeneous transmission potential

Let us rewrite the estimated reproduction ratio in matrix form as

$$R^{\text{estim}}(t) = \frac{I_{\text{tot}}(t+1)}{I_{\text{tot}}(t)} = \mathbf{F}^T \mathbf{R} \mathbf{x}(t), \quad (7)$$

where is $\mathbf{F}$ the unit column vector ($F_i = 1, \forall i$). If we assume that $\mathbf{v}^* = \mathbf{F}^T$ ($v_i^* = 1$), then we can apply R leftwards in equation (7) and get $R^{\text{estim}}(t) = R^{\text{ref}}$ at any time and for any spatial distribution $\mathbf{x}$. Now, the requirement $\mathbf{v}^* = \mathbf{F}^T$ imposes that R is proportional to a left stochastic matrix. Indeed $\mathbf{F}^T R = R^{\text{ref}} \mathbf{F}^T$ means that $\rho_i = \sum_j R_{ji} = R^{\text{ref}}$, so that each column sums to R^{ref} .

Proof of equation (3)

Combining equations (1) and (7), we get the time evolution of the estimated reproduction ratio:

$$R^{\text{estim}}(t) = \frac{\mathbf{F}^T R^{t+1} \mathbf{x}(0)}{\mathbf{F}^T R^t \mathbf{x}(0)}. \quad (8)$$

We insert this into equation (2) and get

$$\Delta(t) = \frac{1}{R^{\text{ref}}} \frac{\mathbf{F}^T (R - R^{\text{ref}}) R^t \mathbf{x}(0)}{\mathbf{F}^T R^t \mathbf{x}(0)}. \quad (9)$$

We introduce the eigenvectors of R (other than $\mathbf{v}$): $\mathbf{w}_\alpha$ is an eigenvector with corresponding eigenvalue Λ_α . Analogously, we define the corresponding dual vector $\mathbf{w}_\alpha^*$. Then, we decompose $\mathbf{F}^T$ in the dual basis: $\mathbf{F}^T = \mathbf{v}^* + \sum_\alpha (\mathbf{F}^T \mathbf{w}_\alpha) \mathbf{w}_\alpha^*$. Using this decomposition in equation (9) and applying R leftwards on the dual eigenvectors we get

$$\Delta(t) = - \frac{\mathbf{F}^T \left[\sum_\alpha \left(\frac{\Lambda_\alpha}{R^{\text{ref}}} \right)^t \left(1 - \frac{\Lambda_\alpha}{R^{\text{ref}}} \right) \mathbf{w}_\alpha \mathbf{w}_\alpha^* \right] \mathbf{x}(0)}{\mathbf{F}^T \left[\mathbf{v} \mathbf{v}^* + \sum_\alpha \left(\frac{\Lambda_\alpha}{R^{\text{ref}}} \right)^t \mathbf{w}_\alpha \mathbf{w}_\alpha^* \right] \mathbf{x}(0)}. \quad (10)$$

The denominator is $C(t)$ in equation (3):

$$C(t) = \frac{1}{\mathbf{F}^T \left[\mathbf{v} \mathbf{v}^* + \sum_\alpha \left(\frac{\Lambda_\alpha}{R^{\text{ref}}} \right)^t \mathbf{w}_\alpha \mathbf{w}_\alpha^* \right] \mathbf{x}(0)}. \quad (11)$$

$C(t)$ is always strictly positive because it is proportional to $\mathbf{F}^T R^t \mathbf{x}(0)$ and tends to $\mathbf{v}^* \mathbf{x}(0)$, that is, the component of the initial condition onto the eigenspace of the Perron eigenvalue. This component is always non-zero because $\mathbf{x}(0)$ is nonnegative (as it is a spatial distribution of infections) and no nonnegative vector can be orthogonal to a strictly positive vector. Thus, the numerator gives the trend and sign of $\Delta(t)$. Equation (10) then gives the value of the factors z_α in equation (3):

$$z_\alpha = -\mathbf{F}^T (\mathbf{w}_\alpha \mathbf{w}_\alpha^*) \mathbf{x}(0). \quad (12)$$

If the eigenvalue is degenerate, one should simply replace $\mathbf{w}_\alpha \mathbf{w}_\alpha^*$ with the appropriate projector over the whole eigenspace. Note that, as discussed before, the denominator in equation (10) is always real and positive so any complex phase of z_α must arise from Λ_α and $\mathbf{w}_\alpha \mathbf{w}_\alpha^*$.

Calculation of τ_α and T_α

We isolate in equation (3) the contribution of each mode $M_\alpha(t)$: $\Delta(t) = C(t) \sum_\alpha M_\alpha(t)$, where

$$M_\alpha(t) = M_\alpha(0) \left(\frac{\Lambda_\alpha}{R^{\text{ref}}} \right)^t = M_\alpha(0) \left(\frac{|\Lambda_\alpha|}{R^{\text{ref}}} \right)^t e^{i\theta_\alpha t} = M_\alpha(0) e^{-t/\tau_\alpha} e^{i\theta_\alpha t}, \quad (13)$$

where we used the definition of τ_α given in Main. The decaying term with characteristic time τ_α is visible.

If Λ_α is real and positive, then $\theta_\alpha = 0$ and the oscillating term vanishes. If Λ_α is real and negative, then $\theta_\alpha = \pi$ and the oscillating term becomes an alternating sign: $e^{i\theta_\alpha t} = (-1)^t$. This is an oscillation with period $T_\alpha = 2$, which is compatible with the definition of T_α given in Main. Finally, if $\Lambda_\alpha \notin \mathbb{R}$, then $\bar{\Lambda}_\alpha$ is also an eigenvalue, where the bar denotes complex conjugation. We will call $\bar{\alpha}$ the index corresponding to that eigenvalue: $\Lambda_{\bar{\alpha}} = \bar{\Lambda}_\alpha$. Also, the projector over the eigenspace of $\Lambda_{\bar{\alpha}}$ is the elementwise complex conjugate of the projector over the eigenspace of Λ_α , meaning that $z_{\bar{\alpha}} = \bar{z}_\alpha$, and, thus, that $M_{\bar{\alpha}} = \bar{M}_\alpha(0)$. Then α and $\bar{\alpha}$ contribute as a pair, as follows:

$$\begin{aligned} M_\alpha(t) + M_{\bar{\alpha}}(t) &= e^{-t/\tau_\alpha} \left[M_\alpha(0) e^{i\theta_\alpha t} + M_{\bar{\alpha}}(0) e^{-i\theta_\alpha t} \right] \\ &= 2 e^{-t/\tau_\alpha} |M_\alpha(0)| \operatorname{Re} e^{i\theta_\alpha t + \phi_\alpha} = 2 e^{-t/\tau_\alpha} \cos \left(\frac{2\pi}{T_\alpha} t + \phi_\alpha \right). \end{aligned} \quad (14)$$

Here we used the definition of T_α given in Main to explicitly show the emergence of the oscillating term with period T_α .

Local over- and underestimation of R^{ref}

The Collatz–Wielandt inequalities tell us that, for any spatial distribution of infections $\mathbf{x}$, $\min_{i|x_i \neq 0} (R\mathbf{x})_i / x_i \leq R^{\text{ref}}$ and $\max_{i|x_i \neq 0} (R\mathbf{x})_i / x_i \geq R^{\text{ref}}$. Given that $r_i^{\text{estim}} = (R\mathbf{x})_i / x_i$, out of equilibrium there will always be at least one community overestimating the reference reproduction ratio ($r_i^{\text{estim}} > R^{\text{ref}}$) and one underestimating it ($r_i^{\text{estim}} < R^{\text{ref}}$).

Reconstruction of the reproduction operator from data

The main data used for the reconstruction of reproduction operators for mainland France are Meta colocation maps, which were calculated from mobile phone data and show the proportion of time that residents of different communities spend in proximity⁴⁵. More precisely, they give the probability p_{ij} that a randomly chosen person who is resident of community i and a randomly chosen person who is a resident of community j are both in the same $600 \text{ m} \times 600 \text{ m}$ square during a randomly chosen 5 min time window in a given week. The data were provided at the resolution of departments (ADM 2). To reconstruct R from these data, we assumed that the expected number of secondary infections generated among the residents of community i by a person who is a resident of community j is given by $R_{ij} = C p_{ij} n_i$, where n_i is the population of spatial patch i and C is an overall transmissibility parameter. Notably, although the value of the spectral radius of R clearly depends on C , the left and right Perron eigenvectors $\mathbf{v}$ and $\mathbf{v}^*$ do not but depend solely on the data. To compute $\mathbf{v}$ and $\mathbf{v}^*$ from R , we computed its right ($\mathbf{v}$) and left ($\mathbf{v}^*$) eigenvectors associated with the largest eigenvalue using the Python function `numpy.linalg.eig` in `numpy v.1.12`. The diagonal elements p_{ii} of the colocation maps quantify the mixing within each community. In a previous study we found that they overestimate it in densely populated areas and when people spend a lot of time at home⁴³. Using that same methodology, we, thus, corrected those entries using Meta movement range maps (section ‘Data availability’). Movement range maps give the daily fraction sp_i of residents of community i that do not leave their $600 \text{ m} \times 600 \text{ m}$ home tile. The probability $p_{ii}^{(sp)}$ of detecting colocations among people staying in their separate homes and, thus, not actually mixing is, in community i ,

$$p_{ii}^{(sp)} = \frac{(sp_i d_i A)(sp_i d_i A - 1) m_i}{n_i(n_i - 1)}, \quad (15)$$

where d_i is the population density in community i , $A = 0.36 \text{ km}^2$ is the area of a single tile, m_i is the number of tiles occupied by community i and n_i is the resident population. We then discounted $p_{ii}^{(sp)}$ from p_{ii} to remove spurious colocations. We show in Supplementary Fig. 2 the magnitude of this for the central week of the grey period of Fig. 1. For most departments, it ranged between 1% and 10%, exceeding the latter value only in the densely populated area of Paris and its surroundings.

Epidemic simulations

The model of epidemic spread used in our simulations is a stochastic discrete-time metapopulation model in which spatially distinct communities are linked through mobility^{26,27,66,67}. We used a synthetic population based on census data from the National Institute of Statistics and Economic Studies in France. We divided this population into 94 spatial communities corresponding to the departments of mainland France except Corsica. Meta colocation maps⁴⁵ and movement range maps were used to reconstruct the coupling p_{ij} between communities i and j and the within-community i mixing p_{ii} . We used a SEIR compartmental model with ϵ as the rate of the E-to-I transition, μ the rate of the I-to-R transition and β the overall transmission rate. The values of ϵ and μ were taken from ref. 68 (COVID-19-like), and β was set so that $R^{\text{ref}} = 1.5$. Following ref. 45, the rate of force-of-infection to which those in community i are subjected is $\beta \sum_j p_{ij} I_j$, where I_j is the number of infected in j . We used a discretization time step $\Delta d = 1$ day. Supplementary Section 1.4. describes in detail each step of the stochastic simulation model. There, we also compute the reproduction ratio for our model according to the next-generation matrix formalism⁶⁹, obtaining

$$\text{NGM}_{ij} = \frac{\beta}{\mu} p_{ij} n_i, \quad (16)$$

and $R^{\text{ref}} = \text{spectral rad}(\text{NGM})$ (Supplementary Section 1.4). Again, n_i is the population of community i .

Data availability

Meta Colocation Maps, which were used to reconstruct the reproduction operators and to infer between- and within-community mixing for stochastic simulations can be requested at <https://dataforgood.facebook.com/dfg/tools/colocation-maps>. Meta Movement Range Maps used to correct within-community colocations are available at <https://data.humdata.org/dataset/movement-range-maps>. Hospital admission data in France are available at <https://www.data.gouv.fr>. French census data can be found at <https://www.insee.fr>. Shapefiles for French departments are available at <https://www.data.gouv.fr/en/datasets/carte-des-departements-2-1/>. All websites were accessed in November 2023.

Code availability

The code used in this study is available here: <https://github.com/ev-modelers/birello-surveillance> (ref. 70). A continuously developed and maintained library allowing us to implement our corrections to estimates of the reproduction ratio from surveillance data can be found at <https://github.com/ev-modelers/rt-from-surveillance> (ref. 60).

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Author contributions

E.V. conceived and designed the study. E.V. and M.R.F. developed the theory. P.B. developed the code and the publicly available library, performed the numerical simulations and analysed the results. B.W. analysed the surveillance data and reconstructed infections from them. P.B., M.R.F., B.W., V.C. and E.V. interpreted the results. E.V. drafted the article. P.B., M.R.F., B.W., V.C. and E.V. contributed to and approved the final version of the article.

Competing interests

The authors declare no competing interests.

Additional information

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